Machine Learning: Netflix Movies and TV Shows Clustering

1. 📌 Project Description

- **Context:** This project focuses on analyzing a dataset of TV shows and movies available on **Netflix as of 2019**, sourced from **Flixable**.
- **Problem Statement:** In 2018, reports highlighted a significant shift: while the number of TV shows on Netflix had nearly **tripled** since 2010, the number of movies had decreased by more than **2,000 titles**.
- **Main Objectives:**
  - Perform **Exploratory Data Analysis (EDA)** to uncover catalog patterns.
  - Understand content diversity across different countries.
  - Identify if Netflix is pivoting toward episodic content.
  - Use **Text-Based Clustering** to group similar content for recommendation potential.

2. 📊 Dataset Overview

- **Data Source:** Flixable (third-party Netflix search engine).
- **Data Size:** The dataset contains **7,787 records and 12 columns**.
- **Key Features:** show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, and description.

3. 📑 Understanding Variables

- **Numerical Variables:** release_year and derived temporal features like year_added, month_added, and day_added.
- **Categorical Variables:** type (Movie/TV Show), title, director, cast, country, rating, genres (renamed from listed_in), and description.

4. 🛠️ Data Wrangling

- **Duplicate Handling:** Checked for redundant records; no duplicates were found in the dataset.
- **Missing Values:**

    - **Director (30.68%)**, **Cast (9.22%)**, and **Country (6.51%)** had high missing percentages and were imputed with **"Unavailable"** labels.
    - Minor nulls in date_added and rating were dropped to maintain data integrity.
- **Outlier Detection:** Used the **IQR (Interquartile Range) Method** on the release_year variable. This identified extreme historical values, which were then capped or filtered to prevent the model from being skewed by rare, very old titles.

5. 🧪 Feature Engineering

- **Temporal Extraction:** Converted date_added to a datetime object to derive separate features for **Year, Month, and Day** content was added.
- **Duration Processing:** Split the original duration column into duration_minutes for movies and seasons for TV shows.
- **Standardization:** Renamed listed_in to **genres** for better clarity and ensured all numerical year features were consistent **int64** types.

6. 🔍 Exploratory Data Analysis (EDA)

- **Content Split:** Movies account for **69.14%** of the catalog, while TV shows make up **30.86%**.
- **Growth Trends:** 2019 was a peak year, with nearly **27.71%** of the total library added in that year alone.
- **Strategic Shift:** Observations confirm that by 2020, Netflix began releasing more TV shows annually than movies, signaling a pivot toward episodic, "binge-worthy" content.

7. 📈 Data Visualization

- **Univariate:** Analyzed the distribution of content types and audience ratings, identifying **TV-MA** and **TV-14** as the most common.
- **Bivariate:** Explored the relationship between producing countries and content volume; the **United States** leads globally, followed by **India** for movies and the **UK** for shows.
- **Multivariate:** Used **Heatmaps** and **Pair Plots** to identify correlations (or lack thereof) between release timing and duration.

8. 🎯 Hypothesis Testing

- **Hypothesis 1 (T-Test):** Confirmed a significant difference in **Duration** between Movies and TV Shows.

- **Hypothesis 2 (ANOVA):** Found a significant association between **Content Rating** and **Release Year**.
- **Hypothesis 3 (Chi-Square):** Confirmed a significant relationship between the **Producing Country** and the **Type** of content (Movie vs. TV Show).

9. ⚙️ Data Pre-processing

- **NLP Pipeline:** Applied to text-heavy columns (description and title).
- **Cleaning Steps:** Performed contraction expansion, lowercasing, and removal of punctuations, URLs, digits, and stopwords.
- **Normalization:** Used **WordNetLemmatizer** to reduce words to their root forms and applied **Part of Speech (POS) Tagging** to maintain semantic context.

10. ✂️ Feature Manipulation & Selection

- **Vectorization:** Used **TF-IDF Vectorizer** (1000 features) to convert text into a numerical matrix.
- **Selection Pipeline:** Applied VarianceThreshold, SelectKBest (Mutual Information), and **Recursive Feature Elimination (RFE)** to isolate the most predictive features for classification.

11. ⚖️ Data Transformation & Scaling

- **Standardization:** Applied **StandardScaler** to all numeric features.
- **Verification:** Statistical checks confirmed the transformation resulted in a **mean of 0 and a standard deviation of 1** for all features, ensuring model stability.

12. 📉 Dimensionality Reduction and Data Splitting

- **PCA:** Reduced the high-dimensional TF-IDF data while retaining **95% of the variance**.
- **Clustering:** Used **K-Means Clustering** to segment the catalog into **5 distinct thematic clusters**.
- **Splitting:** Performed an **80/20 Stratified Split** to ensure training and testing sets had a balanced distribution of these clusters.

13. ⚖️ Handling Imbalanced Data

- **SMOTE:** Applied **Synthetic Minority Over-sampling Technique** to balance the training data across all clusters. This prevents the machine learning models from becoming biased toward the larger content groups.

14. 🤖 ML Model Implementation

Three classifiers were implemented to predict cluster membership:

- **Logistic Regression:** Achieved **100% accuracy** (baseline).
- **Random Forest:** Delivered **99.08% accuracy**, showing robustness for non-linear patterns.
- **XGBoost:** Selected as the final model with **99.83% accuracy**.

15. 🏆 Model Optimization

- **Hyperparameter Tuning:** Used **GridSearchCV** with 5-fold cross-validation.
- **Parameters:** Optimized C and solver for Logistic Regression, and learning_rate and n_estimators for XGBoost to ensure the highest possible generalization score.

16. 🏁 Conclusion

- **Final Model Selection: XGBoost** was chosen for production due to its superior accuracy, ability to handle non-linear data, and high performance across **Precision, Recall, and F1-Score**.
- **Business Impact:** The clustering provides a framework for **Enhanced Recommendations**, allowing Netflix to suggest "thematic gems" to users.
- **Strategy Insights:** Analysis proves Netflix has transitioned into a **Global TV Powerhouse**, strategically tailoring its library (like Indian Dramas) to regional market preferences.